Information Theory + Evolution = Everything

Universal Novelty Generation Algorithm

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September 12, 2025

Abstract

Evolution explains persistence and extinction through selection but struggles to explain the emergence of genuine novelty. We identify the missing mechanism: a third operation beyond reinforcement and elimination that generates new information states through resonance. Using cellular automata as implementation proofs, we demonstrate how discrete systems access continuous potential to create unprecedented states. This operation, combined with integration and dissolution, appears identically across all scales from physical to biological to semantic. The framework extends evolution to its complete form: active generation of novelty through wave interference at specific phase relationships.

Keywords: Information theory; Evolution; Complexity; Cellular automata; Emergence; Wave mechanics; Universal computation; Novelty generation

1 Introduction

Darwin's theory of evolution revolutionized our understanding of complexity by showing how simple processes generate diverse forms through variation and selection. Natural selection elegantly explains why successful patterns persist and unsuccessful ones disappear. But the theory has an acknowledged gap: where does genuine novelty come from?

Natural selection theory can only select from existing variation, it explains the survival of the fittest but not the arrival of the fittest. The standard model attributes novelty to error plus chance reinforcement—random mutations that happen to prove beneficial. While a fairly satisfying explanation, this mechanism alone does not fully explain the consistent emergence of increasing complexity, the convergent evolution of similar solutions in independent lineages, or the acceleration of evolutionary innovation over time.

Meanwhile, physics has increasingly recognized information as fundamental to reality: from quantum information theory to digital physics, evidence mounts that reality is computational at its base. If information is the substrate, then *evolution must operate on information itself*, not just on biological forms.

This paper identifies the missing piece: a third operation beyond persistence/extinction that generates genuine novelty through resonance between existing states. When discrete states meet at specific phase relationships, wave interference generates genuinely new frequencies that emerge from the interaction. This resonance or creation operation operation, combined with Integration (persistence) and Dissolution (extinction), completes our understanding of how complexity emerges at every scale.

Evolution provides the selection mechanism—explaining why patterns persist or perish while Information theory provides the operations—how patterns interact and transform. Combined, they form a complete system: information operations generate possibilities, evolution selects among them, and critically, the selected operations can generate genuinely new operations. This recursion—evolution selecting information processes that themselves evolve—drives complexity generation.

2 The Limitation of Binary Systems

To understand the necessity of the third operation, we begin with Conway's Game of Life [3]. This cellular automaton demonstrates infinite complexity emerging from simple rules, while revealing the fundamental limitation of systems restricted to predetermined states.

2.1 Conway's Game of Life

In 1970, mathematician John Conway created a universe on a grid. Each cell is either alive (black) or dead (white), with eight neighbors. The entire grid updates simultaneously each generation according to three rules:

- A dead cell with exactly three living neighbors becomes alive— modeling reproduction where the right density of life creates new life.
- A living cell with two or three neighbors stays alive—modeling survival with adequate resources.
- All other cells die— death from isolation or overcrowding.

These simple rules generate infinite complexity: patterns oscillate, emit gliders, build structures. The Game of Life is *Turing complete*—it can compute anything computable.

2.2 The Fundamental Constraint

Conway's Game observes an absolute limitation: no new states ever emerge. A cell that starts as black or white can only toggle between black and white. The patterns grow arbitrarily complex through rearranging the same two states that existed from the beginning.

The limitation runs deeper than it appears. Conway's Game can represent infinite different configurations through combinations of its two states—just as binary computers can represent any computable number or pattern through zeros and ones. The Game can encode and process any information that can be discretized into binary form.

The crucial distinction lies between **representing states and generating fundamentally new states**. Conway's Game operates within a closed state space: it creates infinite combinations of black and white, yet remains unable to access anything beyond these two states. The system is ontologically closed—no computation within the Game can create "gray" or "red" or any state undefined at initialization.

Conway's Game is a piano with only two keys. You can play infinitely complex rhythms and patterns with those two notes, even encode entire symphonies through temporal patterns. Yet you cannot generate a third frequency.

3 A Different Game

Now consider a game where the states themselves evolve:

3.1 Initial Conditions

At $T = 0$, instead of a grid with black and white cells, we have a uniform field. it isn't empty nor full—these are distinctions. Instead, this field is undifferentiated: no boundaries exist to create information, no gradients to enable flow, no distinctions to allow meaning.

We will call this initial condition *gray*—representing undifferentiated potential. Between 0 and 1 exist infinite numbers, between black and white exist infinite shades. This gray contains infinite possibility compressed into perfect uniformity. As Shannon established, a uniform signal carries zero information [4]—information requires distinction.

3.2 The First Distinction

At $T = 0 + \epsilon$, the system performs signal comparison between neighboring regions. Perfect uniformity is impossible—thermal noise, quantum fluctuations, or computational rounding create tiny variations. When neighboring regions compare signals and find differences, these differences amplify through feedback.

Small variations appear, the perfect gray develops slight gradients—some regions marginally denser, others marginally less. These variations amplify through feedback. Like supercooled water crystallizing when disturbed, the continuous gray approaches a critical threshold where smooth gradients can no longer be sustained.

At $T = 1$, the system undergoes *phase transition*. The continuous spectrum snaps into discrete states: black (0) and white (1). The universe now has its first information—the ability to encode this versus that, here versus there, yes versus no. This symmetry breaking transforms infinite potential into finite distinction.

3.3 The Three Operations

Once black and white exist as discrete states, three operations emerge from wave mechanics whenever oscillating states interact:

Operation 1 - Integration (Consonance): Similar states encounter each other and their waves interfere constructively. *Black regions surrounded by black become more stably black*—peaks align with peaks, amplifying the signal. This creates persistence and structure.

Operation 2 - Dissolution (Dissonance): Opposite states meet and their waves interfere destructively. *Black and white at a boundary cancel out*—peaks meet troughs, neutralizing the signal. This enables change and adaptation.

Operation 3 - Creation (Resonance): States meet at specific phase relationships where waves generate new frequencies through nonlinear interaction. Black and white in resonance produce gray—a new frequency emerging from their interaction. This gray exists at a specific point in the continuous spectrum between 0 and 1.

When black and white create a shade like $gray_{0.37}$, they access the continuous potential that existed before discretization and pull through something new. The difference between **shuffling a deck and printing a new card.**

Creation doesn't emerge after black and white exist; their emergence IS *creation or discretization*. This first distinction described in the previous section necessarily contains within itself what will become Integration (establishing stable states) and Dissolution (boundaries between them).

There is a recursive trap here: discrete states must exist for operations to be applied, and creating discrete states is the first operation. **It's absolutely impossible to operate directly on the undifferentiated continuous, a state must be selected, made discrete.**

What we're calling operation 3 contains all future operations. Every subsequent Creation—black and white generat-

ing gray, gray generating further shades—follows the same pattern established in this first moment.

3.4 To Understand Recursiveness

The game expands through a simple rule: anything created through Operation 3 performs all three operations. When $gray_{0.37}$ emerges from black and white in resonance, $gray_{0.37}$ can integrate with itself (forming stable $gray_{0.37}$ regions), dissolve when meeting incompatible states, and create new states when meeting other grays, black, or white in resonance.

Each creation adds a new player. $gray_{0.37}$ meeting white in resonance generates $gray_{0.82}$. $gray_{0.82}$ meeting black generates $gray_{0.15}$. The palette expands following specific patterns—certain resonances are easier to achieve, certain states more stable once created.

Run this game forward and complexity explodes, open-ended complexity where the state space itself evolves. New types emerge at any point, limited only by the system's ability to *find resonance conditions*.

The expanding palette demonstrates the system's range. Between black (0) and white (1) exist infinite possible grays—continuous gradations. Each represents a distinct state that emerged through creation (or discretization) operations, accessing different points in the continuous potential.

This grayscale capability means the system can generate *any grayscale image through appropriate initial conditions and evolution time*. This cellular automata running on grayscale demonstrates just the single-channel version, but it can execute across multiple channels simultaneously (see section 5).

Since the gray game can produce any grayscale or colored configuration through multiple-channels, it can produce this one:

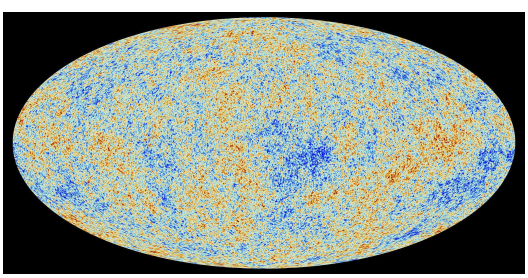

Figure 1: The Cosmic Microwave Background—grayscale intensity variations of 0.001% from the mean, representing our universe's oldest observable structure [2].

Our first picture from when the universe became transparent at 380,000 years old. We visualize it with color where blue represents slightly cooler (denser) regions and red represents slightly warmer (less dense) regions. This picture contains every structure, every planet, every star and every pattern that existed or will ever exist in our universe, including this conversation.

The CMB represents one possible output among infinite possibilities: the game that generates infinite grays between black and white, that could produce any photograph or measurement or pattern, could happen to produce our universe's oldest observable structure when run with our quantum fluctuations as input.

4 The Physical and Mathematical Correspondence

These three operations eerily map to fundamental principles in physics and math:

Operation 1 (Integration) = Gravity = Addition = Persistence: The universal attractive force pulls matter together, creating structure and stability through constructive interference, just as gravity creates persistence through mutual attraction, and addition combines quantities into larger wholes. Masses integrate into larger structures—planets, stars, galaxies—through like attracting like.

Operation 2 (Dissolution) = Entropy = Subtraction = Extinction: The second law of thermodynamics drives systems toward more probable states, dissolving organized patterns into heat through destructive interference, just as subtraction removes and reduces, making room for new configurations.

Operation 3 (Creation) = Phase Transition = The Equals Sign in E=mc² = Reproduction: Einstein's equation declares energy and matter are the same thing in different forms, with the equals sign represents the transformation operation itself, the resonance that allows one state to become another. When matter becomes energy in fusion, or energy becomes matter in particle creation, creation manifests at the fundamental level. In mathematics, this is creating new variables or sets through declaration, in physics, it's phase transitions where continuous fields become discrete particles.

When you say "apple," you create a boundary between apple and not-apple from continuous reality, discretizing through declaration, creating a new set that didn't exist before the naming.

This might seem counterintuitively simple, but if energy equals matter ($E=mc^2$), *and* entropy drives energy (second law), *and* energy drives evolution (selection for efficient energy use), *and* particles are also waves (wave-particle duality), *therefore* we're simply **applying evolutionary principles to wave mechanics.**

Evolution is how physics selects which patterns persist when waves interact and these three operations are the only possible outcomes: they must either reinforce (constructive interference), cancel (destructive interference), or generate new frequencies (nonlinear interaction).

This is simply mathematical and physical necessity and appears at every scale, while evolution selects for systems that successfully perform and balance all three operations to dapt to changing conditions.

No additional mathematical equations are needed here because wave interactions are well understood and the game is simple, applying recursively. The complexity emerges when context is inserted and you ´re looking at specific states.

These three operations suggest a pattern while actual implementations require contextual detail, whether calculating protein folding or predicting market dynamics. Just as Einstein's $E=mc^2$ provides the understanding while the field equations handle actual calculations, the value here lies in recognizing the common mechanism.

4.1 The Edge of Chaos

When thinking about the system dynamics another scale-invariant property emerges from these systems: they consistently operate at *the edge of chaos* [5], the critical region where resonance operations become maximally probable.

Systems need enough stability for patterns to maintain phase relationships, yet enough fluctuation for those patterns to achieve resonance and generate new frequencies. Too much order locks the system into Integration-Dissolution cycles that merely rearrange existing states, while too much chaos destroys information before resonance can occur, adaptation requires a form of *unstable equilibrium*.

The critical point between these regimes is where water exists as both ice and liquid at 0°C, where forest densities enable fire percolation without total burnout, where heart rhythms exhibit 1/f variability between rigid periodicity and arrhythmia, and where evolution proceeds through mutation rates poised at error catastrophe.

This criticality seems necessary for the third operation to emerge, considering its ubiquity across natural information-processing systems. Biology discovered this through selection pressure, eliminating both hyperstable systems that couldn't adapt and hypervariable ones that couldn't preserve beneficial traits.

5 Universal Video Decoder

We can see how evolution converged on this solution when analyzing the evolution of our visual apparatus.

5.1 The Meaning of Color

Vision is the gray game repeated across multiple channels, which means evolution had to discover the same mathematical solution we derived from first principles through billions of years of selection pressure eliminating whatever wasn't optimal.

The grayscale system we derived works for any wavelength range because detecting something automatically creates its opposition through absence—a cell that fires for red photons necessarily doesn't fire for not-red, creating meaning from presence versus absence exactly as our black-white game does. This fundamental property of detection itself means we can build color vision by running parallel games recursivelly across different wavelength ranges, with each channel creating its own opposition (black/white, red/green, yellow/blue)

Color adds a different layer of meaning, like the photograph of an apple tree: grayscale shows form—branches, leaves, fruit shapes, but then add color and new information emerges: red fruit are ripe, weather appears through sky color, the tree color tells you about its health and the season. The same grayscale structure carries different meaning depending on its color encoding, *color adds semantic layers about state, time, and condition.*

In our universe, white light contains all wavelengths superposed, much like our undifferentiated gray. When white light passes through a prism, the interaction reveals red, orange, yellow, green, blue, violet, *it becomes the operator* that differentiates the continuous spectrum into the discrete bands we can name.

5.2 To See a World

Evolution faced the same challenge: how to extract meaningful information from the continuous electromagnetic spectrum using biological hardware that must operate through discrete detection events.

The simplest organisms began with basic photosensitive proteins that could only detect presence or absence of light, enabling them to distinguish day from night and orient toward or away from light sources. These primitive eyespots implemented our single-channel gray game in the most minimal biological hardware possible, just clusters of cells that changed chemically when struck by photons.

The first color discrimination emerged when some of these photosensitive cells developed sensitivity to specific wavelengths rather than all light equally. Most early vertebrates found the two-channel solution when S-opsins emerged in their existing vision system, creating cells that fire for short wavelengths (blue). When luminosity cells detect light but blue cells don't, the organism perceives the longer wavelenghts (yellow/orange/red) by complementarity.

Most mammals still use this two-channel system today because it provides enough information for their survival needs without the metabolic cost of additional detector types. Dogs see the world in blues and yellows because this distinction, combined with luminosity information, suffices for navigation, hunting, and predator avoidance.

Primates developed the three-channel system through gene duplication around thirty million years ago when the L-opsin gene copied itself and one version accumulated mutations that shifted its peak sensitivity, creating two distinct variants we now call L and M opsins. This created what we now call red and green receptors from what was originally a single type, suddenly enabling discrimination within the longer wavelengths that previously all appeared as "yellow." Red fruit became visible against green foliage for the first time, providing such enormous survival advantage that this mutation swept through primate populations.

Three receptors represent the mathematical minimum for complete color encoding: luminosity plus two spectral samples at opposite ends of the spectrum would suffice (like red+blue giving you green+yellow as not-red+not-blue). But biology requires four because rods and cones operate in completely different light regimes — rods for night vision, cones

for daylight.

Evolution kept both because survival requires seeing in starlight and sunlight, not just optimal laboratory conditions. Human vision now uses **four types of photoreceptors**: rods that work in dim light detecting only brightness, plus three types of cones (RGB) that work in bright light detecting different wavelength ranges. These four inputs generate six perceptual variables through opponent processing: black/white, red/green, yellow/blue.

Three cone types provide the minimum for reconstructing any spectral distribution through linear combination—the mathematical principle behind RGB color spaces in digtal screens. Combined with rods for night vision, these four photoreceptor types generate six perceptual variables through opponent processing: black/white from luminosity, red/green from L versus M cone comparison, and blue/yellow from S cones versus the combined L+M signal.

Now this gives us a system that can represent any RGB image across time—any movie or video—and if it and can represent any possible video it can represent any visual universe within those wavelength bounds, and this is precisely what evolution converged upon through selection pressure—not specialized detectors for specific objects or patterns but an *universal visual decoder,* the minimum configuration capable of representing any visual information.

Natural selection eliminated everything more complex as wasteful and everything simpler as insufficient, leaving only the mathematical optimum that we can derive from first principles and that biology implements in our retinas.

Furthermore, Berlin and Kay's linguistic studies [8] provide cultural confirmation of this mathematical and biological sequence, documenting how every human language develops color terms in almost the same order:

- **Black** and white emerge first as the fundamental luminosity distinction
- **Red** follows because long-wavelength discrimination provided immediate survival value in identifying ripe fruit and blood
- **Green** and yellow develop as agriculture makes medium-wavelength discrimination culturally important
- **Blue** appears last despite S-opsins being evolutionarily ancient because fine discrimination of short wavelengths rarely mattered for terrestrial survival where the sky is simply blue or not blue.

Tetrachromats with a fourth cone type see real colors the rest of us cannot, but these colors remain unnamed because language requires social consensus. Evolution converged on three color channels because three *provided all the survival-relevant distinctions*. The fourth channel adds real information that simply doesn't matter enough to propagate through the population or the language.

The recursive application of our initial grayscale came across parallel channels simultaneously seems to be the software our visual system is applying, with each channel implementing the presence-absence game at different wavelengths while their interactions at critical phase relationships generate the full spectrum of color experience

6 Everything Everywhere All at Once

This "game" and its three operations appear in every information processing system, at every scale:

- **Quantum**: Particles form Bose-Einstein condensates (integration), measurement collapses superposition (dissolution), vacuum fluctuations generate virtual particle pairs (creation)
- **Molecular**: Similar molecules crystallize into lattices (integration), incompatible substances disperse (dissolution), catalysts enable novel reactions (creation)
- **Cellular**: Proteins aggregate into structures (integration), enzymes break down nutrients (dissolution), endosymbiosis created mitochondria through bacterial-archaeal resonance (creation)
- **Neural**: Repeated firing strengthens synapses (integration), unused pathways prune (dissolution), disparate patterns connect in insights (creation)
- **Social**: Shared practices become traditions (integration), outdated systems collapse (dissolution), different ideas combine into innovations (creation)
- **Cosmic**: Gravity binds matter (integration), entropy drives expansion (dissolution), gas clouds collapse into stars (creation)

Each scale implements the identical pattern at different timescales and substrates, a fundamentally informational universe computes its own evolution through these same three operations everywhere, with complexity emerging from context—the same principle by which chess's simple rules produce explosively high combinatory power

7 Completing Evolution

Some of the mechanisms behind evolution are more adequately explained by this model:

Convergent evolution produces nearly identical solutions in independent lineages. Camera eyes evolved separately in vertebrates, cephalopods, and cnidarians, each achieving the same optical principles through different developmental pathways. Random mutation should produce random solutions, yet evolution consistently discovers the same resonance points.

Different lineages find identical solutions because certain resonance points in morphological space are optimal for specific functions. Just as all cultures independently discover the wheel for transportation and the arch for spanning gaps,

evolution repeatedly discovers camera eyes, echolocation, and powered flight because these represent stable peaks where biological materials achieve resonance with physical constraints. The solutions are determined by here physics and biology can successfully interact, optimal resonance points that peaks in morphological space—like multiple climbers finding the same mountain summit from different starting points.

Punctuated equilibrium shows long stasis interrupted by rapid innovation bursts. Stasis represents systems *locked in integration-dissolution cycle*s, existing patterns reinforcing or eliminating each other without accessing new possibilities.

Innovation bursts occur when environmental shifts enable creation operations, systems achieving resonance and creating new states opens entirely new regions of possibility space. The Cambrian explosion exemplifies this: new developmental genes enabled resonance between previously incompatible body plans, generating unprecedented morphological diversity in geological moments

Accelerating complexity emerges naturally from recursive expansion. Prokaryotes performed limited operations until endosymbiosis created eukaryotes with new resonance possibilitiesm which enabled multicellularity with expanded interaction space. Multicellularity then enabled nervous systems operating at entirely new frequencies, each level enables more combinations and thus rapid innovation than the last. Functional leaps requiring multiple coordinated changes become comprehensible through resonance: flight evolution required simultaneous modifications to bones, muscles, metabolism, and behavior.

A single developmental resonance can reorganize entire body plans, just as one frequency shift changes all overtones in a harmonic series. Evolution makes jumps that incremental mutation cannot account for because resonance generates coordinated leaps.

8 The Geometry of the System

The system generates consistent geometric signatures across scales, the mathematical consequences of entropy-driven processes in different dimensions. The mathematical and geometrical formalisms underlying these operations are developed in detail in [6], but their manifestation in evolutionary systems deserves mention here.

One-dimensional projection produces Gaussian distributions through the central limit theorem. When independent random variables sum together, their distribution converges to a bell curve regardless of the underlying individual distributions. The Cosmic Microwave Background follows an almost perfect Gaussian with temperature variations of 0.001% from the mean because it represents countless quantum fluctuations summed together.

Two-dimensional projection generates branching through *Brownian motion* creates dendritic patterns via diffusion-limited aggregation—particles wandering randomly stick where they first make contact, building fractal trees. Rivers, lightning, trees, neurons, and blood vessels all result from this same process.

Three-dimensional projection produces spheres through surface minimization. For a given volume, the sphere has minimum surface area, making it the lowest energy configuration under isotropic pressure. Soap bubbles, stars, cells, and planets become spherical because physics minimizes energy expenditure.

Adding time rotates these patterns into spirals through angular momentum conservation. Systems that grow while rotating naturally produce logarithmic spirals—galaxies, hurricanes, nautilus shells. The golden ratio emerges in biological systems because it optimizes packing efficiency during rotational growth.

These geometries—Gaussian distributions, dendritic branching, spherical forms, and spiral growth—arise from basic physical principles operating at different dimensional scales. The three operations generate these specific patterns because they represent energy minimization solutions in their respective dimensions.

This completes our framework: the operations provide the mechanism, the edge of chaos provides the optimal operating regime, and these geometric signatures show the physical manifestation of information processing in spacetime. Evolution operates through these mathematical constraints, leaving characteristic traces at every scale.

9 Synthesis and Discussion

This might seem like overgeneralization, but let's start from our basic premise: we say the substract is information. Then we have to define it, what exactly is information? Information theory defines it through probability distributions (entropy), Landauer shows it needs physical substrate (energy-matter), and we get multiple theories that ties information and matter together (It from Bit, IIT, QIT).

The definition of matter also depends on the definition of energy, which depends on the definition of entropy, which could be viewed either as a logical trap or an insight. But just by saying "this is made of information", we immediately set *how it is shaped* (the geometry of its space) and *what constrains its behavior* (entropy).

Then the follow up question would be: what does information do? At the simplest form and across all these explanations, information behaves as waves— either in physics (eletromagnetic waves, gravitational waves, quantum mechanics) or information processing (signal processing).

And when waves interact, only three outcomes are possible: consonance, dissonance and resonance. This gives a complete generative system, constructive interference maintains persistence and structure, destructive interference enables change and non-linear interaction generates the novelty.

This gives us both aspects of reality: how to *BE* (information crystallized as matter, maintaint patterns through integration against entropy) and how to *BECOME* (information flowing as energy, transforming through the operations).

9.1 The Evolution of Time

One last question remains, why does information (or time) move unidirectionally? The equations allow bidirectional flow, but the substrate itself doesn't.

Information requires difference— that's standard information theory— a uniform signal carries no meaning. So for information to exist, there must be distinct states, something must differ from something else. Now, if information could flow backwards, the act of comparison would corrupt what it's comparing against.

If future states could alter past states, *nothing could maintain memory or any form of persistence*, not even matter. It's a requirement for causality and the very definition of entropy, it's the primitve restriction for any pattern to exist. **Time flows in one direction because it's required to maintain coherence**, at every scale.

How this ties to evolutionary principles through persistence/extinction so far has been pretty evident, but one distinction still remains between physics and biology here: how living beings differ from non-living matter by creating their own perpendicular arrow of time.

We can observe nature's also found a similar solution with the **Central Dogma**, the unidirectional information flow from DNA to protein creating a parallel solution to the arrow of time, and thus providing conditions to maintain embodied coherence.

It's a necessary solution that must have evolved for systems to predict and adapt in this informational environment, with major evolutionary transitions (prokaryotes to eukaryotes, individuals to societies) implementing new layers of nested unidirectional information flows.

Therefore, by simply declaring what it is (information), we constrain what it does (wave operations), how it's structured (information geometry) and where it flows (entropy)— the complete generative system from a single premise. These are the four irreducible primitives that generate and govern every system or phenomena we can observe.

10 Conclusion

The undifferentiated gray we began with is simultaneously everything and nothing—the Ruliad [1], the set containing all sets, God, the singularity, and heat death. It is Russell's paradox resolved through process: instead of a static set trying to contain itself (impossible), we have continuous potential that becomes discrete through the act of signal checking, identity checking.

Genuine novelty emerges through resonance, completing Darwin's mechanism for both the survival and arrival of the fittest. By recognizing that information, geometry, entropy, and wave operations form irreducible primitives, we see evolution and physics as complementary descriptions of the same process—information transforming itself under physical constraints.

More detailed mathematical formalisms, geometric proofs and other specifics can be found in my other paper "The Geometrical Theory of Communication: How Information Becomes Meaningful Through Algebraic Operations" [6], with computationally tractable approaches like 'Observer Theory" [1] and 'The Holy Trinity of Information" [7] providing implementation instructions beyond this conceptual game.

Every phenomenon examined follows these rules without exception across all scales of reality. We invite rigorous examination of these claims, and specially falsification—finding a logical inconsistecy, an edge case or pattern where this doesn't apply or a simpler explanation than the four stated primitives.

Science advances through such challenges, but personally — and I acknowledge many have been wrong on this account before — these appears to be rules of our chess, everything else is combinations from now.

This does not solve every problem, far from it, instead it might provide new and more precise solutions and much work still remains translating and applying these principles and identifying these patterns in their specific domains and contexts.

Persistence and survival still requires a tremendous amount of energy from humanity, the implications and aplications of these principles that are slowly being uncovered through convergent discovery still needs extensive work, and more importantly how our third operation — resonance — manifests across these scales.

I might add an appendix later showing how Euler's Identity, Schroedinger's time-dependant equation and general relativity emerge from the observed primitives, but the rigorous or curious reader can ask their preferred LLM and apply it to their context. Instead, I'd rather argue a true theory of everything is not a physics-exclusive one, but applies across all domains of science, including what we now consider as "metaphysical".

When two beings interact, maintain balance between integration and dissolution and resonate to create something new—reproduction, adaptation, love— is the operation that needs further articulation, and where science and civilization are heading to maintain coherence and survival.

This might sound like magical non-sense, grandiose or maybe an oversimplification to some, but to others it's becoming obvious in hindsight, these are all old pieces of the puzzle and the picture is becoming ever clearer now, but the details still need to be adjusted.

There are still many islands to map in this newly discovered continent, it's a continuous process and I'm certainly not the first to arrive here: the artists and mystics are our first explorers. The paradigm shift, and science must now develop the rigor to match their wisdom.

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